

MirAI School of Technology: Capstone

Project Directory & Evaluation Matrix

This document serves as the official directory for the final B.Tech Streamlit & AI Capstone Projects. It contains an exhaustive list of 30 real-world problem statements designed to test your mastery of Streamlit UI components, Git version control, Data Visualization, and Gemini AI integrations (including Vision and Audio capabilities).

Evaluation Rubric (100 Points)

To identify top performers for potential internship referrals or showcase opportunities, projects will be evaluated strictly on the following major hackathon parameters.

Evaluation Category	Max Points	Scoring Criteria
1. Technical Implementation & Architecture	25	Flawless execution of Python code. Proper use of st.session_state to prevent memory loss, st.form to optimize API calls, and clean data pipelines using Pandas DataFrames. Zero terminal errors during runtime.
2. AI Integration & Prompt Engineering	20	Advanced use of the Gemini API. Does the app utilize system prompts, f-strings for dynamic context, and multimodality (Camera Input / Mic Recorder)? The AI must act as a tailored engine, not just a generic chatbot.
3. UI/UX & Data Visualization	20	Professional dashboard aesthetic. Must use column layouts, expanders, and

Evaluation Category	Max Points	Scoring Criteria
		dynamic KPI cards (st.metric with deltas). Includes interactive elements like st.data_editor or st.map to present data beautifully.
4. Deployment & Cloud Engineering	15	Successfully deployed and live on Streamlit Community Cloud, Render, or Hugging Face Spaces. The requirements.txt file must be perfectly configured without local system dependencies.
5. Open-Source Branding (GitHub)	10	The GitHub repository must feature a customized, terminal-style README.md explaining the architecture, setup instructions, and a link to the live app.
6. System Design & Documentation	10	Creation of a clear system architecture diagram (e.g., using Mermaid or Lucidchart) and a concise technical design document explaining the data flow, API integration strategy, and logic modules.

The 30 Real-World Problem Statements

Select one of the following domains. You may pivot or combine ideas, but the core technical requirements remain the same.

Category A: FinTech, Quants & Economics

- **1. The Expense Roaster:** A dashboard where users upload a mock CSV of their monthly expenses. Gemini analyzes the data and brutally roasts their discretionary spending, providing a strict budget recovery plan.
- **2. Hyper-Local Inflation Tracker:** Users log the price of basic grocery items (milk, eggs, rice) over a month. The app plots the inflation trend via `st.line_chart` and uses AI to predict the next month's budget requirement.
- **3. Crypto Sentiment Terminal:** A financial dashboard that takes recent news headlines as input, uses AI to classify the sentiment (Bullish/Bearish), and updates a live `st.metric` UI card accordingly.
- **4. The Subscription Auditor:** Users input all their monthly digital subscriptions. The app visualizes the annual cost drain and uses Gemini to suggest cheaper, open-source, or bundled alternatives.
- **5. Salary Negotiation Simulator:** A voice-driven app (using the mic recorder) where the AI plays the role of a strict HR manager. The user practices negotiating a 96,400 INR salary, and the AI grades their persuasion tactics.

Category B: EdTech & Campus Survival

- **6. Roommate Agreement Generator:** A highly interactive `st.form` where students set quiet hours, guest rules, and chore splits. Gemini generates a legally-sounding, humorous contract that can be downloaded.
- **7. B.Tech Jargon Translator:** An app designed for non-technical parents. Students type in computer science jargon, and the AI translates it into everyday analogies (e.g., explaining APIs as a restaurant waiter).
- **8. Voice-Notes to Flashcards:** Users speak a chaotic lecture summary into the microphone. The app transcribes the audio and commands Gemini to generate a structured study guide and 5 quiz questions.
- **9. The "Will I Get Caught?" Plagiarism Checker:** Users paste their essay and the source material. Gemini acts as an academic integrity board, generating a percentage

score and highlighting risky paraphrasing.

- **10. Campus Event Heatmap:** A dashboard using st.map. Users input coordinates of various college events. The app visualizes where the most activity is happening and uses AI to suggest optimal networking routes.

Category C: Health, Fitness & Wellbeing

- **11. Hyper-Local Macro Engine:** A fitness app where users input a specific daily protein target and a strict budget. Gemini generates a meal plan utilizing localized grocery items to hit the exact macronutrients.
- **12. "Roast My Form" Analyzer:** Users utilize st.camera_input to take a photo of their starting posture for a pushup or squat. The Gemini Vision model analyzes the biomechanics and highlights flaws.
- **13. Sleep Debt Visualizer:** A dashboard tracking mock sleeping hours over two weeks. It calculates the cumulative "sleep debt" and uses AI to map out a precise recovery schedule.
- **14. The Fridge-to-Feast Generator:** Users snap a photo of the random ingredients left in their fridge. The AI identifies the items and outputs a step-by-step recipe to prevent food waste.
- **15. Gym Routine Randomizer:** An interactive st.data_editor where users maintain a database of exercises they know. The app randomly selects 5 exercises, generating a unique, balanced workout of the day (WOD).

Category D: Productivity & Enterprise Automation

- **16. Voice-Activated Excel Macro Builder:** A tool for tax professionals. The user speaks their desired spreadsheet action (e.g., "Filter Q3 revenue"), and the app generates the exact Python Pandas or Excel VBA code.
- **17. The AI Resume Critic (Tech-Roast):** Users paste their resume text and a target job description. The AI acts as a ruthless Silicon Valley recruiter, highlighting missing keywords and weak bullet points.
- **18. Desk Setup Rater:** Users submit a photo of their workstation via webcam. The AI evaluates the ergonomics, cable management, and aesthetic, assigning a productivity score.
- **19. Cold Email Generator:** An app utilizing shareable URLs (st.query_params). Users input a company name and role, and the AI drafts a highly personalized networking email to send to recruiters.

- **20. Meeting Action-Item Extractor:** Users paste a chaotic raw transcript from a 1-hour meeting. The AI parses the text into an interactive `st.data_editor` table assigning tasks, deadlines, and owners.

Category E: Lifestyle, Entertainment & Gaming

- **21. Spoiler-Free Lore Guide:** An interactive library for thriller or fantasy novels. Users select their current chapter, and the AI answers questions about the plot strictly without revealing future events.
- **22. The Fan-Zone Sports Commentator:** A dashboard tracking live-simulated cricket scores. The AI acts as a highly biased team commentator, generating dynamic, hype-filled text based on the `st.metric` fluctuations.
- **23. 1v1 Code Clash Arena Lobby:** Utilizing URL state management, this app generates a unique match ID. Two players join the same URL to receive the exact same AI-generated algorithmic puzzle.
- **24. Aesthetic Book Matchmaker:** Users describe a specific "vibe" or aesthetic (e.g., "dark academia in the rain"). The AI recommends three novels that match the mood, bypassing standard genre classifications.
- **25. Interactive Visual Novel Engine:** A text-based RPG where users speak their actions into the microphone. The AI acts as the Dungeon Master, updating the story and generating the next scenario dynamically.

Category F: Accessibility & Advanced Utilities

- **26. Digital Life-OS (Wellbeing Tracker):** A comprehensive dashboard that visualizes screen time data and uses Gemini as a holistic lifestyle coach to suggest physical replacement activities.
- **27. Visual-to-Text Signage Explainer:** Users upload photos of confusing parking signs or complex public transport maps. The Gemini Vision API breaks down exactly what the sign means in plain language.
- **28. Local Volunteer Matchmaker:** A map-based app where local NGOs are plotted. Users input their specific skills (e.g., coding, heavy lifting), and the AI matches them to the nearest organization in need.
- **29. The "Explain it Like I Play" Dictionary:** Users select a favorite video game (like Minecraft or Valorant) and input a complex engineering topic. The AI explains the concept entirely through the mechanics of that specific game.
- **30. Audio-Accessibility Communicator:** An app designed for individuals with temporary

vocal loss. It features a dashboard of massive, touch-friendly buttons that trigger AI-generated text-to-speech phrases for emergency or daily communication.

Submission Instructions

All projects must be committed to your main branch by 25th August at 11:59 PM. Ensure your GitHub README contains the live deployment link. Be prepared to defend your architectural choices and prompt engineering strategies during the final review. A LinkedIn post tagging MirAI School of Technology is a mandatory requirement for your submission to be considered.